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AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors: US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - Chapter V

**76.9% global operational AI
compute = USA**

**1.59x energy cost EU/US (PPA-
adjusted)**

**3.46:1 CACI ratio US/EU (Power
Mode)**

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CHAPTER V

Prospective Scenarios 2026-2030

This chapter constitutes the heart of the original contribution of this study. By applying the methodological protocol defined in Chapter II (2x2 matrix, six divergence metrics, CACI calibration), we construct four scenarios for the evolution of the transatlantic relationship in AI, energy, and semiconductors for the 2026-2030 period. Each scenario is determined by the combination of two critical uncertainties identified in Chapter III: the degree of US protectionism and the European strategic response capacity. We then evaluate each scenario on its six metrics, before synthesising the tipping point conditions between trajectories.

5.1 Predetermined elements: what will not change

In accordance with the Schwartz method (1991), we distinguish predetermined elements (quasi-certain trends by the 2030 horizon) from critical uncertainties (factors whose evolution depends on political decisions not yet taken). Four predetermined elements structure all scenarios.

EP1 - Exponential growth in AI compute demand. Semiconductor sales doubled in two years (2023-2025), the power of installed AI chips doubles every seven months (Epoch AI), and no sign of slowing down is observable as of February 2026. Even under the hypothesis of a deceleration in scaling laws (Chinchilla saturation), the diffusion of AI towards inference, robotics, and autonomous agents will maintain a strongly growing compute demand.[1]

EP2 - Persistent concentration of compute in the United States. The US/EU ratio of 17.6:1 in raw installed compute (Chapter III, April 2026 dashboard snapshot: 2,759,968 vs 156,632 PFLOP/s), resulting in a CACI Power Mode ratio of 3.46:1 once weighted by the geometric formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$, reflects investment decisions taken in 2022-2025 whose effects materialise until 2028-2029 (data center construction delays: 18-36 months). Even an immediate political reversal would not alter the installed stock before the end of the decade.

EP3 - Growing energy tension. Global data center consumption, estimated at 415 TWh in 2024, will reach 800-950 TWh by 2030 according to IEA projections (Chapter III). The asymmetry in energy costs (US 1.4-1.7x cheaper after PPA correction, 2-3x cheaper on unadjusted industrial Eurostat tariffs) will persist, unless massive investment in European nuclear power, whose deployment delays (SMR: 5-7 years for the first reactors) exceed the 2030 horizon.[2]

EP4 - Section 232 regulatory framework in place. Proclamation 11002 of January 14, 2026, is a legal fait accompli. Unlike IEEPA tariffs (annulled by the Supreme Court on February 20, 2026[3]), Section 232 tariffs rest on a confirmed legal basis. The Secretary of Commerce's report on the data center semiconductor market is expected by July 1, 2026, and may recommend an extension or modification of tariffs. Regardless of the direction taken, the legal instrument will remain available.[4]

5.2 Critical uncertainties and 2x2 matrix

5.2.1 Axis 1: Degree of US protectionism

The first uncertainty concerns the evolution of American policy between two poles. The moderate pole corresponds to maintaining the January 2026 status quo: a 25 percent tariff limited to re-exported advanced chips, broad domestic exemptions, a US-EU trade agreement capping semiconductor tariffs at 15 percent, and no significant extension to cloud or models. The EU core (France, Germany) remains in the 'trusted partners' category. The aggressive pole assumes an extension after the July 2026 report: tariffs extended to derived semiconductors and equipment, GPU quotas for the EU (including France), restrictive conditions for access to frontier AI cloud, and the use of compute as a trade negotiation lever (compute-for-concessions).[5]

5.2.2 Axis 2: European strategic response capacity

The second uncertainty concerns the EU's ability to deploy a coherent and rapid response. The reactive pole corresponds to fragmented national responses, dispersed investments, an AI Act creating additional compliance costs, and slow deployment of AI Factories/Gigafactories (bureaucratic delays, 24+ month authorisations). The proactive pole assumes the accelerated implementation of the AI Continent program (19 AI Factories plus up to 5 Gigafactories of 100,000+ GPUs), the adoption of Special Compute Zones (180-day authorisation), the effective mobilisation of the InvestAI fund (20 billion EUR), and the pooling of French nuclear capacity as a competitive advantage.[6]

5.2.3 Matrix and scenario naming

The intersection of these two axes produces four scenarios. Reactive EU + moderate US protectionism gives Scenario A (Reinforced Status Quo, slow drift towards dependence). Reactive EU + aggressive US gives Scenario B (Digital Fracture, structural European decoupling). Proactive EU + moderate US gives Scenario C (Asymmetric Partnership, junior Western technological partner). Proactive EU + aggressive US gives Scenario D (Contested Sovereignty, race for autonomy under pressure).

5.3 Scenario A - Reinforced Status Quo (Moderate protectionism + Reactive EU)

5.3.1 Narrative

Following the July 2026 report, the Secretary of Commerce recommends maintaining the 25 percent tariff on re-exported advanced chips but without extending it significantly. The August 2025 US-EU trade agreement is respected: semiconductor tariffs for the EU remain capped at 15 percent.[7] The EU, reassured by this status quo, slows the deployment of its own initiatives. EuroHPC AI Factories struggle to reach nominal capacity (authorisation delays, inter-state coordination). Gigafactories are postponed to 2029-2030. The InvestAI fund is partially mobilised (8-10 billion EUR out of 20). European companies continue to rely heavily on US cloud, whose performance and costs remain unbeatable.

5.3.2 Metric trajectories

M1 - Compute ratio (US/EU installed GPUs): goes from 17.6:1 raw (2025) to 18-22:1 raw (2030) on operational installed compute. The gap widens slightly as US investments accelerate (Stargate, xAI mega-clusters, Meta) while the EU only adds the 19 AI Factories (max 25,000 GPUs each, approx. 475,000 public GPUs, an order of magnitude below a single US hyperscaler).[8]

M2 - FLOP cost gap (EU/US): remains in the 2.4-3.2x range. The absence of aggressive tariffs on the EU maintains access to US cloud at prices close to current levels, but European energy costs continue to weigh in.

M3 - US cloud share in European AI spending: goes from 70 percent (2024) to 72-75 percent (2030). European providers (OVHcloud, Deutsche Telekom) maintain their 15 percent in the sovereignty segment but do not gain ground on generative AI services.

M4 - AI productivity (percent/year): US +2.5-3.0; EU +1.0-1.5. The EU realises part of the AI potential via downstream applications (SAP, Siemens, fintech), but slow adoption and compute deficit cap the gains.

M5 - Energy dependence (data center TWh): EU approx. 115 TWh in 2030 (+65 percent vs 2024). French nuclear power absorbs part of the demand, but the absence of Special Compute Zones delays the network connection of new data centers.

M6 - CACI(US)/CACI(EU): goes from 3.46:1 (April 2026) to 4-5:1 (2030). The gap widens moderately as the F factor (compute) accumulates on the US side while E costs (energy) weigh on the EU side.

5.3.3 Consequences for France

This scenario is the most probable in the short term (estimated probability: 40-50 percent). It is also the most insidious: the absence of a visible shock demobilises European actors, while dependence deepens structurally. French companies benefit from access to US cloud for AI adoption (BNP Paribas, Airbus, TotalEnergies via AWS/Azure), but this adoption reinforces the lock-in described in Chapter IV. The AI productivity deficit relative to the United States (-1.0 to -1.5 points per year) accumulates over five years, widening the competitiveness gap by 5 to 8 GDP points.

5.4 Scenario B - Digital Fracture (Aggressive protectionism + Reactive EU)

5.4.1 Narrative

The July 2026 report leads to a significant extension. The Secretary of Commerce recommends tariffs extended to semiconductor equipment and derived products, with a tariff offset program reserved for companies investing in American production.[9] The 15 percent EU agreement is revised upwards or accompanied by restrictive conditions (volume quotas on advanced GPUs, reciprocity requirements on the AI Act). Simultaneously, access to frontier AI cloud is made conditional for non-US entities (API access limitations to frontier models, weight restrictions). The EU, fragmented, fails to formulate a coherent response: member states divide between accommodation (Nordic countries, Netherlands) and confrontation (France, Italy).

5.4.2 Metric trajectories

M1 - Compute ratio: goes from 17.6:1 raw (2025) to 25-35:1 raw (2030). GPU quotas limit European imports at the moment demand explodes. AI Factory projects are compromised by the inability to procure Nvidia/AMD GPUs at the planned volumes.

M2 - FLOP cost gap: jumps to 4-6x. Extended tariffs, combined with quotas and energy asymmetry, massively increase European compute costs. French companies face a 3x to 5x surcharge for model training.

M3 - US cloud share: paradoxically rises to 78-82 percent. Failing a credible local alternative, European companies wanting access to frontier AI must go through US hyperscalers, at the price conditions they dictate. Sovereign services (OVHcloud, Scaleway) lack the hardware to offer competitive GenAI services.

M4 - AI productivity: US +2.5-3.5; EU +0.3-0.8. European AI potential is severely constrained. The McKinsey Global Institute estimates that with slow adoption, European productivity would not exceed 0.3 percent, close to stagnation.[10]

M5 - Energy dependence: EU approx. 95 TWh only (2030), not by virtue but by default - the lack of GPUs limits data center construction. Ironically, the compute constraint mitigates the energy constraint.

M6 - CACI ratio: explodes from 3.46:1 (April 2026) to 6-8:1 (2030). This is the scenario where the gap is largest, with all three CACI factors deteriorating simultaneously on the European side: F capped by quotas, E inflated by tariffs, L weakened by an accelerated brain drain to the United States.

5.4.3 Consequences for France

This scenario (estimated probability: 15-20 percent) represents the worst-case. France suffers structural technological decoupling: compute-intensive projects (Mistral foundation models, Exotec robotics, Dassault simulations) are relocated to the United States or depend on increasingly costly US cloud access. Time-to-market for French AI solutions lengthens by 25 to 40 percent. SMEs, unable to absorb surcharges, renounce frontier AI and opt for degraded solutions (smaller open-source models, local inference). The cumulative productivity gap with the United States reaches 10 to 15 points over five years.

5.5 Scenario C - Asymmetric Partnership (Moderate protectionism + Proactive EU)

5.5.1 Narrative

US protectionism remains moderate (as in A), but the EU exploits this window to accelerate its own investments. AI Factories are deployed on schedule (2026-2027), the first 100,000+ GPU Gigafactories are ordered end-2026 and delivered in 2028.[11] France plays a central role thanks to its nuclear fleet (65-70 percent of electricity mix, competitive marginal cost), and Special Compute Zones are designated on former industrial sites with heavy network connections.[12] However, the EU de facto accepts a status as a junior technological partner: it uses Nvidia/AMD GPUs (no European AI ASIC design champion), depends on TSMC/Samsung/Intel foundries for production, and its foundation models remain one step below US leaders.

5.5.2 Metric trajectories

M1 - Compute ratio: drops from 17.6:1 raw (2025) to 8-10:1 raw (2030) on installed compute. Gigafactories and private investment (InvestAI plus industrial co-investments) add 1-2 million H100 equivalents in Europe, narrowing the gap without closing it.

M2 - FLOP cost gap: drops to 1.5-2.0x. French nuclear power and Gigafactory economies of scale compress energy and infrastructure costs, although a residual gap persists (absence of proprietary GPU design).

M3 - US cloud share: drops slightly to 60-65 percent. European sovereign services gain market share in regulated segments (defense, health, finance), while US cloud retains the majority of commercial workloads. The market segments into sovereign and performance.

M4 - AI productivity: US +2.5-3.0; EU +1.8-2.5. The EU reaches 60-80 percent of its theoretical potential thanks to sufficient local compute for large-scale adoption of downstream applications, even if frontier model training remains dependent on US hardware.

M5 - Energy: EU approx. 140 TWh (2030). Demand is higher than in A because European compute increases, but nuclear and planned SMRs absorb most of it. RTE France confirms the feasibility of +10 GW subject to network investments.

M6 - CACI ratio: drops from 3.46:1 (April 2026) to 2.0-2.5:1 (2030). This is the most favourable realistically achievable scenario at the 2030 horizon. The F factor improves significantly, E benefits from nuclear, but L remains slightly lower (the US AI ecosystem being more attractive for top talent).

5.5.3 Consequences for France

This scenario (estimated probability: 15-20 percent) is the most favourable for France in the short-to-medium term. France becomes the EU's AI energy hub thanks to its nuclear fleet, attracting data center and Gigafactory investments. French companies gain access to competitive local compute for inference and fine-tuning, reducing dependence on US cloud for standard uses. Mistral and French startups can train specialised models locally. However, frontier model training remains dependent on US hardware, and strategic autonomy is partial: France is sovereign in application, but not in the creation of fundamental technologies.

5.6 Scenario D - Contested Sovereignty (Aggressive protectionism + Proactive EU)

5.6.1 Narrative

US protectionism intensifies (as in B), but the EU responds with determination. The American threat becomes the political catalyst for an unprecedented European industrial mobilisation since the AIRBUS project of the 1970s. The AI Continent program is accelerated and extended: the 5 Gigafactories are ordered as an emergency, France announces 20 GW of nuclear capacity dedicated to AI data centers by 2032 (combining fleet extension, new EPR2s and SMRs), the DARE project (European RISC-V) is escalated to design AI accelerators reducing dependence on Nvidia.[13] Simultaneously, the EU negotiates alternative technological alliances (Japan, South Korea, Taiwan) to secure GPU and foundry supplies.

5.6.2 Metric trajectories

M1 - Compute ratio: evolves from 17.6:1 raw (2025) to 12-15:1 raw (2030). The EU invests massively but starts from very far. US quotas slow imports, but alternative alliances and local production (Gigafactories using Samsung/Intel GPUs as alternatives to Nvidia) partially compensate.

M2 - FLOP cost gap: 2.5-4.0x initially (2027, peak tariff shock), then progressive reduction towards 1.8-2.5x (2030) as Gigafactories scale up and GPU alternatives mature.

M3 - US cloud share: drops to 50-55 percent (2030), the most pronounced decline of the four scenarios. Geopolitical defiance and US restrictions push European companies towards sovereign alternatives, even if imperfect. US hyperscalers lose ground in regulated segments.

M4 - AI productivity: US +2.5-3.5; EU +1.2-2.0. The EU traverses a productivity trough in 2027-2028 (transition period where US restrictions bite but European investments are not yet operational), then a partial catch-up from 2029.

M5 - Energy: EU approx. 150-160 TWh (2030). This is the most energy-intensive scenario for the EU, with massive local data center construction creating huge demand. French nuclear power becomes a continental strategic asset, but network pressure is at its maximum.

M6 - CACI ratio: follows a U-shaped trajectory: degradation from 3.46:1 to 8-12:1 in 2027-2028 (peak shock), then improvement to 4-7:1 by 2030. The result depends heavily on European execution speed: each year of delay on Gigafactories prolongs the period of maximum vulnerability.

5.6.3 Consequences for France

This scenario (estimated probability: 15-20 percent) is the most ambitious and riskiest. It places France at the heart of an unprecedented European technological sovereignty effort. Massive nuclear investments (SMR, fleet extension) become a top-tier geopolitical issue. The DARE/RISC-V project could, if successful, constitute the first credible European alternative to Nvidia GPUs for AI, but on a 5-7 year horizon, well beyond 2030. In the short term (2026-2028), France traverses a period of maximum vulnerability where surcharges and shortages degrade competitiveness, before a catch-up conditional on infrastructure deployment speed.

5.7 Comparative synthesis and tipping points

5.7.1 Metric synthesis table

Table 11 below consolidates the trajectory of the six divergence metrics for the four scenarios by the 2030 horizon, anchored on the April 2026 dashboard snapshot (US/EU raw operational compute 17.6:1, CACI Power Mode 3.46:1).

5.7.2 Tipping points between scenarios

The real trajectory will likely follow a hybrid path between these scenarios. Three tipping points determine possible transitions.

First tipping point: the July 2026 Commerce report. This report will determine whether US protectionism extends (shift to B or D) or remains targeted (maintenance in A or C). Indicators to watch: evolution of the US semiconductor trade deficit, CHIPS Act fab filling rates (Intel, TSMC Arizona, Samsung Taylor), domestic political pressure (2026 midterms). Phase 1 negotiation results (report due April 14, 2026) will be an early signal.[14]

Second tipping point: EU Gigafactory deployment speed. The Commission expects the first Gigafactories to be operational in 2027-2028. If this schedule is held, the EU shifts towards proactive scenarios (C or D). If authorisation, financing, or hardware supply delays push deliveries to 2029-2030, the EU remains in reactive mode (A or B). The CFG proposal for Special Compute Zones (180-day authorisation vs 24+ months currently) is the key acceleration factor.[15]

Third tipping point: the French decision on nuclear for AI. France possesses a unique asset in Europe: a nuclear fleet providing 65-70 percent of electricity at a globally competitive marginal cost. The decision to dedicate significant capacity (10-20 GW) to AI data centers, via fleet extension, new EPR2s and SMRs, will determine whether France becomes Europe's AI energy hub or loses this position to others (Scandinavia with hydro, Eastern Europe with low land costs). This tipping point is specifically French and determines France's position within the European scenarios.[16]

5.7.3 The convergence point: 2028

The four scenarios converge on a common critical point in 2028. This is the year when: (i) compute demand will exceed installed capacity in Europe, creating hardware bottlenecks (even under moderate protectionism); (ii) the full effects of extended tariffs (if adopted) will be felt; (iii) Gigafactories, if deployed on time, will start producing significant local compute; (iv) data center energy demand will saturate network connection capacity in several member states. The year 2028 constitutes the moment of truth when Europe will discover if it is on an A/B trajectory (growing dependence) or C/D (catch-up started). Decisions taken in 2026-2027 (US Commerce report, Gigafactories, French nuclear) will be irreversibly engaged.

5.8 Origins of the tipping point: juridico-technical foundations

The 'Great Decoupling' scenario is not built ex nihilo. It is the logical projection, at the 2028 horizon, of a control architecture whose foundational layers are already operational in 2026. Identifying these layers is a matter of academic rigour: distinguishing documented trends from extrapolated projections.

5.8.1 The legal layer: extraterritoriality as a structural instrument

The first layer is legal and largely precedes AI policy. The CLOUD Act (Clarifying Lawful Overseas Use of Data Act, 2018) establishes that any cloud provider subject to US jurisdiction is required to produce data regardless of where such communications, records, or other information are stored.[17] This law, confirmed by federal case law (United States v. Microsoft), creates a fundamental dissociation between the physical location of data and its legal nationality.

The immediate consequence for AI compute is radical: an H100 cluster physically located in Dubai, Singapore, or an AWS eu-west-1 center in Ireland remains legally American. If ordered by the US government, the operator (AWS, Azure, Google Cloud) is legally bound to comply, regardless of the client's will or local legislation. Microsoft acknowledged before a French court in 2024 that it could not guarantee data sovereignty for European clients in the event of a legally grounded US injunction.[18]

This pre-existing legal architecture is the substrate on which the Cloud-Nationality Pivot grafts: the Cloud Sovereignty Mandates projected for 2028 do not create a new power. They activate and systematise an existing jurisdictional power by extending it to the operational compute layer.

5.8.2 The technical layer: from location verification to cluster throttling

The second layer is technical. The US AI Action Plan of July 23, 2025, explicitly introduces the concept of 'location verification features' applied to advanced AI chips. Michael Kratsios, OSTP director at the White House, confirmed discussions on software or hardware modifications to enable better location-tracking, explicitly included in the plan.[19]

This announcement is not rhetorical. The BIS already has, since the Framework for Artificial Intelligence Diffusion (January 2025), a compute-tiering control architecture for destination countries (Tier 1-2-3) and compute caps per entity and country.[20] The BIS Affiliates Rule, suspended for one year in November 2025

but maintained in principle, stipulates that affiliation to a parent entity in a restricted country is sufficient to deny access to advanced compute, regardless of physical location.[21]

The technological trajectory towards 2028 is therefore: (i) chips equipped with location verification mechanisms, (ii) automatic reporting to the BIS in case of deviation, (iii) the ability to suspend access or throttle performance via export licence. This is not science fiction: it is the extension to compute of a principle already applied to software (OFAC sanctions, service access freeze).

Critical limitation: Throttling production clusters is technically complex and potentially disruptive for operators themselves. NVIDIA does not currently have a remote deactivation mechanism for H100/B200 GPUs in production. Such a mechanism would require significant architectural modifications to firmware and remote attestation protocols (via TPM or equivalent). The 2028 scenario is therefore conditional on the effective implementation of these modifications over 24-36 months.

5.9 The Cloud-Nationality Pivot mechanism

5.9.1 The trigger: Cloud Sovereignty Mandates as extension of export controls

The 2028 tipping point scenario assumes that the United States crosses a qualitative threshold: moving from hardware access control (BIS chip controls) to operational compute service access control (Cloud Sovereignty Mandates). This threshold is not an arbitrary rupture; it responds to a structural flaw identified in the current control regime.

This flaw is documented: despite BIS restrictions, investigations revealed that approx. 1 billion USD of Nvidia chips reached China via third countries (Malaysia, UAE, Singapore) in the first months of 2025.[22] The AI Action Plan's response (location verification and cluster monitoring) is the first step towards continuous post-export control.

The plausible trigger in 2028 is an executive order extending Framework for AI Diffusion obligations to the cloud layer. It would impose a 'Data Residency and Jurisdiction Compliance' certification on all US hyperscalers operating advanced offshore clusters, with revocation of compute access on US soil as the enforcement mechanism.

5.9.2 Dissociating Physical Factor / Sovereign Factor in CACI

This is where the scenario produces its most significant impact on the CACI model. Under the Cloud-Nationality Pivot, the compute variable $F(r)$ decomposes into two components: $F(r) = F_{\text{phys}}(r) \times F_{\text{sov}}(r)$, where F_{phys} is physically installed compute and F_{sov} is the operational sovereignty factor (fraction of F_{phys} outside US jurisdiction).

Note the distinction between this dynamic 2028 F_{sov} and the static Sovereign CACI introduced in Chapter I (Fig 1.8). The Chapter I Sovereign CACI captures who owns the compute today (April 2026 snapshot). The 2028 F_{sov} captures who controls compute under a hypothetical Mandate regime, which depends on cloud workload share, not installed capacity. The two metrics agree where compute is owned by domestic

operators; they diverge where local clusters are owned by US-side operators (UAE 99.6%, EU cloud workloads majorly on AWS/Azure/GCP).

Calibrated estimations for 2028, under the Mandate activation hypothesis, are presented in Table 12.

5.10 Emergence of jurisdictional AI blocs (2028-2030)

The Great Decoupling does not produce a binary US/non-US world. It produces fragmentation into blocs of variable intensity, according to each zone's capacity to develop credible sovereign compute. Four blocs emerge.

5.10.1 The Extended American Bloc (American AI Alliance)

The July 23, 2025 AI Action Plan explicitly lays the foundations for an American AI Alliance: exporting the full US tech stack (hardware, models, software, standards) to aligned allies in exchange for aligned export controls.[23] The strategy is 'carrot and stick': aligned allies access frontier chips and models without additional restrictions; those who refuse are exposed to Foreign Direct Product Rule mechanisms and secondary tariffs.

Confirmed Tier 1 members: US, UK, Canada, Australia, Japan, South Korea, Netherlands, Germany, France (subject to export control alignment). For these countries, effective F_{sov} increases: their entities access US hyperscalers without restriction, and sovereign compute in development (EU Gigafactories) receives preferential treatment.

5.10.2 The Eurasian Sovereign Bloc

China constitutes the only complete example of a pre-existing sovereign bloc. With an F_{sov} of approx. 0.98 and a national cloud ecosystem (Alibaba, Tencent, Huawei) operating outside US jurisdiction, Cloud Sovereignty Mandates have no direct traction. The Chinese constraint remains the shortage of advanced chips, but the American bloc cannot throttle a Huawei Ascend 910B cluster.

Post-2028 dynamics: China holds the only large-scale sovereign compute outside the American bloc. Countries seeking to emancipate themselves from Mandates face a binary choice: conditional American compute or Chinese compute under other forms of dependence.

5.10.3 Digital Non-aligned: an untenable position

India, Brazil, Southeast Asia, and Gulf countries (without special US treaties) constitute a digital non-alignment bloc. Their position is structurally uncomfortable: too dependent on US hyperscalers to shift to sovereignty, insufficiently integrated into the US alliance to escape restrictions in case of geopolitical disagreement.

The UAE case illustrates this with quantitative force. Fig 1.8 documented that 99.6% of UAE F_{total} is owned by US-side actors (Stargate UAE, Microsoft, OpenAI), collapsing Sovereign CACI from a Physical 55.7 to just 6.0. Dubai invested massively to become an AI hub, notably via AWS/Microsoft/G42 agreements. Yet, G42 was already pressured in 2024 to break ties with Chinese entities as a condition for advanced chip

access.[24] Under Mandates, this pressure becomes systemic: Gulf compute, physically present but legally American, would become a permanent negotiation lever.

5.10.4 The European Bloc: between alliance and autonomy

Europe occupies an intermediate and evolving position. Legally Tier 1, the UE nevertheless maintains a strategic autonomy ambition that the American Alliance does not fully satisfy.

The Cloud and AI Development Act (CADA), expected Q1 2026, attempts to resolve this by defining an 'EU Sovereignty Level' that would structurally exclude CLOUD Act-subject providers from sensitive public procurement.[25] The October 2025 Cloud Sovereignty Framework defines three assurance levels, with SOV-3 requiring the provider to be beyond the reach of non-European extraterritorial legislation.[26]

This architecture is under construction, but its timing is problematic: CADA will be at best operational in 2027-2028, precisely when US Mandates could be activated. The vulnerability window is maximal.

Note from Chapter I: the EU is largely sovereign on installed compute (99.2% of F_{total} is EU-owned). The vulnerability is not on installed F but on the cloud workload layer (the compute actually used by EU firms, majorly on AWS/Azure/GCP). CADA targets exactly this layer.

5.11 Transversal impacts on scenarios A-D

The Cloud-Nationality Pivot superimposes on the four scenarios, modifying their conclusions non-linearly. It adds a third dimension: the degree of installed compute autonomy. Table 13 synthesises the impact.

5.12 Implications for France: the question of genuinely sovereign compute

5.12.1 'Sovereignty washing' as a systemic risk

The term 'sovereignty washing' designates the practice of US hyperscalers marketing sovereign cloud offers by implanting data centers on European soil, while remaining subject to the CLOUD Act.[27] Microsoft acknowledged in its own documentation that it cannot guarantee sovereignty for European clients in case of a legally grounded US injunction.[28]

5.12.2 The French nuclear advantage in the new equation

The sovereign reinterpretation of CACI paradoxically reinforces the French strategic asset. If $F(r) = F_{phys} \times F_{sov}$, the optimal strategy is not just to increase F_{phys} (attracting hyperscalers) but F_{sov} (developing compute independent of US jurisdictions).

French nuclear power creates a top-tier competitive advantage: Special Compute Zones adjoined to carbon-free, competitive nuclear electricity, hosting Gigafactories operated by European entities (OVHcloud, Scaleway, Mistral AI, IONOS), would produce compute with an F_{sov} close to 1, genuinely beyond the reach of Cloud Sovereignty Mandates.

5.12.3 The DARE/RISC-V project: from ambition to strategic necessity

Under Scenario D and especially under Cloud Sovereignty Mandates, the DARE project (Digital Autonomy with RISC-V in Europe, 2025) shifts from a long-term ambition to a strategic necessity.[29] As long as Europe depends exclusively on NVIDIA/AMD GPUs, US control over these architectures creates a residual vulnerability even in European-operated Gigafactories. Genuine strategic autonomy requires the ability to design independent accelerators, an horizon DARE situates at 2030-2032.

5.13 Synthesis: the fourth tipping point

The Great Decoupling is not inevitable. It represents a conditional systemic risk. To the three tipping points in 5.7.2, a fourth is added.

Fourth tipping point: implementation of location verification features in advanced GPUs. If the BIS and US Commerce Department successfully deploy remote attestation mechanisms in H100/B200 chips by end-2026-2027, the technical substrate for the Cloud-Nationality Pivot will be in place. The question will shift from technical feasibility to political decision.

For Europe and France, a controlled decoupling strategy rests on three pillars: (i) accelerating sovereign compute deployment (EU Gigafactories) to increase F_{sov} before Mandates; (ii) securing Tier 1 status in the American alliance; (iii) investing in architectural alternatives (DARE/RISC-V, Huawei Ascend for non-sensitive workloads) to reduce medium-term dependence.

Table 10. 2x2 matrix of prospective scenarios 2026-2030.

Source: author's construction, Schwartz methodology (1991).

	Reactive EU response	Proactive EU response
Moderate US protectionism	Scenario A - Reinforced Status Quo (slow drift towards dependence)	Scenario C - Asymmetric Partnership (junior Western technological partner)
Aggressive US protectionism	Scenario B - Digital Fracture (structural European decoupling)	Scenario D - Contested Sovereignty (race for autonomy under pressure)

Table 11. Comparative synthesis of the four scenarios on the six divergence metrics (2030 horizon).

Source: author's construction; baseline April 2026 snapshot (US/EU raw operational compute 17.6:1, CACI Power Mode 3.46:1).

Metric (2030)	A - Status Quo	B - Fracture	C - Partnership	D - Sovereignty
M1 Raw compute ratio US/EU (operational)	18-22:1	25-35:1	8-10:1	12-15:1
M2 FLOP cost gap	2.4-3.2x	4-6x	1.5-2.0x	1.8-2.5x
M3 US cloud share (pct)	72-75	78-82	60-65	50-55
M4 EU productivity (pct/yr)	+1.0-1.5	+0.3-0.8	+1.8-2.5	+1.2-2.0
M5 EU energy (TWh)	~115	~95	~140	~155
M6 CACI Power Mode ratio	4-5:1	6-8:1	2.0-2.5:1	4-7:1 (post-trough)
Estimated	40-50 pct	15-20 pct	15-20 pct	15-20 pct

probability				
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Table 12. Estimation of F_sov factor by jurisdiction and CACI impact under Cloud Sovereignty Mandates activation (2028).

Source: author's construction; Synergy Research Group (2025), Statista (2025), and Chapter I Fig 1.8 baseline.

Jurisdiction	F_phys cloud-US share	Estimated F_sov	Current CACI (phys baseline)	Post-Mandate CACI impact
United States	~5 pct (domestic cloud unaffected)	1.00	100 (reference)	100 (unchanged)
EU (France, Germany)	~77 pct (AWS/Azure/GCP dominance)	0.28	28.9 (Power Mode)	30-50 pct collapse on workloads
UAE (Dubai hub)	~88 pct (US hyperscaler dominance)	0.12	55.7 phys / 6.0 souv	60-80 pct collapse - illusory hub
Singapore	~82 pct (US hyperscaler dominance)	0.18	high - APAC hub	55-75 pct collapse
China	~2 pct (Alibaba/Tencent/Huawei Cloud)	0.98	15.7 (chip shortage caps)	Unchanged - sovereignty effective
India	~60 pct (AWS/Azure + local)	0.40	22.2 (Power Mode)	Moderate collapse - intermediate pos

Table 13. Impacts of the Cloud-Nationality Pivot (Cloud Sovereignty Mandates 2028) on the four 2x2 matrix scenarios.

Source: author's construction.

Scenario	Without Mandates	With Mandates 2028	EU CACI Impact	Strategic Reading
A - Status Quo	Slow dependence, CACI 4-5:1	Partial activation - hyperscalers cooperate without major restriction	Moderate degradation 15-25 pct on workloads	More stable but security illusion revealed
B - Fracture	Structural decoupling, CACI 6-8:1	Maximal activation - conditional US cloud and restricted chips simultaneously	Double scissor effect: rare chips + conditional compute. CACI ratio potentially > 8:1	Absolute worst case - chips + cloud combo
C - Partnership	Partial catch-up, CACI 2.0-2.5:1	Sovereign Gigafactories absorb shock if EU F_sov rises to 0.45-0.55	Limited impact if deployed on time - Gigafactories = sovereign hedge	Best resilience - prior investment proves its value
D - Sovereignty	Catch-up under pressure, CACI 4-7:1 post-trough	Mandates become political catalyst - accelerate EU deployment and JP/KR/TW alliances	Accelerated U-curve - 2028-29 trough, faster catch-up	Paradoxical: Mandates can accelerate EU sovereignty if response is fast enough

Notes

1. Epoch AI (Jan 2026), Trends in AI Hardware and Compute. Doubling every 7 months combines 1.6x/yr quantity and 1.6x/yr performance. Even a slowdown to 12 months implies quadrupling by 2030.
2. IEA (Apr 2025), Energy and AI, Paris. 800-950 TWh projections correspond to median/high scenarios. Energy ratio (1.4-1.7x PPA-adjusted) derived from dashboard: USA 85, EU 135 USD/MWh.
3. US Supreme Court (Feb 20, 2026), Learning Resources Inc. v. Trump. 6-3 decision: 'IEEPA does not authorize the President to impose tariffs.'
4. White House (Jan 14, 2026), Proclamation 11002, sec (2): Secretary update on data center semi market due July 1, 2026.
5. Tax Foundation (Feb 2026). Aug 2025 US-EU agreement caps semi tariffs for EU at 15 percent, but Proclamation 11002 allows broader tariffs after Phase 1.
6. European Commission (2025), AI Continent Action Plan. Target: triple EU data center capacity in 5-7 years. 19 AI Factories, 5 Gigafactories. InvestAI: 20 bn EUR. CFG (Oct 2025) Special Compute Zones.
7. Tax Foundation (Feb 2026), op. cit.
8. EuroHPC JU (2025). 19 AI Factories approx 475,000 public GPUs, vs single xAI Colossus (200k H100 extensible).
9. Proclamation 11002, tariff offset program section.
10. McKinsey Global Institute (May 2024). 0.3 percent corresponds to slow adoption scenario.
11. EU Council (Dec 2025). Provisional schedule: operational Gigafactories in 2027-2028.
12. CFG (Oct 2025), 'Tripling the EU Data Centre Stock with Special AI Compute Zones.'
13. EuroHPC JU (Mar 2025), DARE project (Digital Autonomy with RISC-V in Europe).
14. Proclamation 11002, sec (2): April 14, 2026 negotiation report due.
15. CFG (Oct 2025). EU DC authorisation currently 24+ months (vs 6-12 US). SCZ would reduce to 180 days.
16. RTE (2024), Futurs energetiques 2050. France +10 GW AI demand by 2030. Nuclear 65-70% of mix.
17. CLOUD Act, Pub. L. 115-141 (Mar 23, 2018), Title III, sec 103(a). 18 U.S.C. sec 2713.
18. Microsoft France, Tribunal judiciaire de Paris (2024). Cited in The Register (Dec 22, 2025).
19. Michael Kratsios (OSTP), Seoul (Aug 2025), cited in TechResearchOnline.
20. BIS, Framework for Artificial Intelligence Diffusion, Fed. Reg. vol 90, no 10 (Jan 15, 2025).
21. BIS, 'Suspension of the Affiliates Rule for One Year' (Nov 10, 2025).
22. Financial Times (July 25, 2025), '\$1bn Nvidia chips smuggled to China'.
23. White House, EO on Promoting the Export of the American AI Technology Stack (July 23, 2025).
24. US Dept of Commerce, G42/Microsoft agreement (2024). Confirmed by Sec Gina Raimondo.
25. European Commission, 2026 Work Programme, CADA, expected Q1 2026.
26. European Commission, Cloud Sovereignty Framework (Oct 2025), SOV-1 to SOV-3.
27. Cristina Caffarra, Eurostack Foundation, cited in The Register (Dec 22, 2025).
28. See note 18.
29. EuroHPC JU, DARE project (Digital Autonomy with RISC-V in Europe), launched Mar 2025.

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Public dashboard: <https://moOogly.github.io/America-First-IA/dashboard/>

Repository: <https://github.com/moOogly/America-First-IA>

AI for Americans First - Fabrice Pizzi - Chapter V